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The Usage of Extended Kalman Filter to Increase Navigation Accuracy of Mobile Units in Closed Spaces

Adilzhan Kereyev K.[1*], Sabyrzhan Atanov K.[2], Timur Toleuov Zh.[3]
[1*]K.Zhubanov Aktobe Regional University, Kazakhstan, adiljan@mail.ru
[2]L.N. Gumilyov Eurasian National University, Kazakhstan, atanov@mail.ru
[3]K.Zhubanov Aktobe Regional University, Kazakhstan

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***Abstract* — This paper presents a navigation system for confined spaces. The system identifies the location of the mobile device by connecting INS (Inertial Navigation System) data with RSSI (Received Signal Strength Indicator) data. The navigation system uses two types of data: the data obtained from the INS is more accurate over a short period of time, but the probability of error increases over time, at which point the RSSI based position area estimate is considered limited positioning accuracy. As a result, the Kalman filter has improved the accuracy of the system. The concept of building an indoor navigation system using data from a Wi-Fi wireless network is considered, the algorithms,mechanisms, and technological aspects of the system implementation are described, and the implementation of such a system is demonstrated. Algorithms of internal navigation systems are proposed. The system has been implemented and tested with promising results.**

Keywords* — *received signal strength indicator; indoor navigation; inertial navigation system; Kalman filter; iBeacon.

I. INTRODUCTION

In the last years, the problem of internal navigation has become really interesting for researchers, because knowing the location (vehicle, robots) is valuable information for various applications.

In addition, the interest in internal navigation is growing due to the fact that technologies intended for external use cannot be used in closed areas. For example, GPS (Global Positioning System) accuracy is lower in buildings because satellite navigation signals cannot pass through roofs, walls, or floors [1],[2]).

The capabilities of satellite navigation systems are well described in [3]. However, GPS might be adapted for operation in buildings but it has restrictions like Increasing of measurement time, degradation of accuracy, cost improvement [4]-[7]. In door, satellite signals are practically inaccessible, therefore, for navigation purposes, special indoor positioning systems, the so-called Indoor navigation systems, are being developed. Due to its great commercial prospects, this direction is becoming more and more popular and has already attracted the attention of such large companies in the market as Google, Apple, Qualcomm, Broadcom, Sony. One way of organizing an indoor positioning system is by placing beacons that periodically send connection packets. The receivers can be different mobile devices with installed applications. For the successful implementation of the internal positioning system, it is necessary to determine the required number of beacons, solve the problem of their placement and the method of determining the location

II. MATERIALS AND METHODS

Positioning the location using RSSI. In this case, the navigation is based on iBeacon sensors, and the localization method is based on the trilateration algorithm. This algorithm uses the distances that are calculated by the RSSI as input. The environment can drown out radio signals (noise, reflection), so the accuracy of the RSSI method is lower than that of others. To get a system that provides the desired location, the RSSI data was combined with the INS sensor date.

I this paper the system of calculation of 2D descrete is considered, where 4 Bluetooth iBeacon modules are presented in the form of quadrangle.

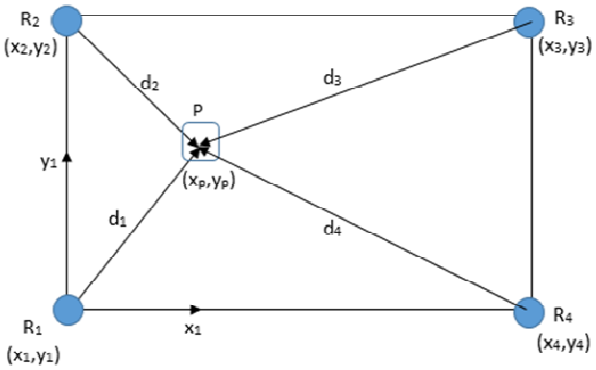


Figure 1. The distance based on RSSI positioning

To identify the location of an object P the follow hypothesis is required [1]:

- iBeacon $(R_1,...,R_4)$ major nodes are locked;
- $(x_i, y_i), i = \{1,\dots 4\}$ coordinates are locked;
- the P node receiver has to receive data from 4 major nodes.

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The distance between the positioning point P and the object can be defined via RSSI value [1]:

$$P_r(d) = P_0 + 10n_p \log_{10}(d_0 / d) \quad (1)$$

P_r – signal strength at a distance of nodes d,
P_0 – signal strength obtained at a distance of nodes d_0,
n_p – distribution constant of a signal.
Equation (1) leads to:

$$P_r(d) = -10n_p \log_{10}(d) + A \quad (2)$$

The distance d can be obtained from the follow relation:

$$d = 10^{A-P_r/10n_p} \quad (3)$$

The distance d_i between the receiving node $P(x_p,y_p)$ and major nodes $R(x_i,y_i)$ is expressed through the follow relation:

$$d_i^2 = (x_i - x_p)^2 + (y_i - y_p)^2 \quad (4)$$

The equation (4) may be formed in this way:

$$d_i^2 = (x_i^2 + x_p^2 - 2x_i x_p + y_i^2 + y_p^2 - 2y_i y_p \quad (5)$$

At this point the equation (5) is concerned with another control point k:

$$d_k^2 = x_k^2 + x_p^2 - 2x_k x_p + y_k^2 + y_p^2 - 2y_k y_p \quad (6)$$

If we substract equation (5) from equation (6), then we obtain the following relation:

$$d_i^2 - d_k^2 + x_k^2 + y_k^2 - x_i^2 - y_i^2 = \\ = 2(x_k - x_i)x_p + 2(y_k - y_i)y_p \quad (7)$$

If we consider $i = 1$ and the variable index $k = \{2, \ldots 3\}$, then we obtain the following system of equations:

$$\begin{bmatrix} 2(x_2 - x_1) & 2(y_2 - y_1) \\ 2(x_3 - x_1) & 2(y_3 - y_1) \\ 2(x_4 - x_1) & 2(y_4 - y_1) \end{bmatrix} \begin{bmatrix} x_p \\ y_p \end{bmatrix} = \\ = \begin{bmatrix} d_1^2 - d_2^2 + x_2^2 + y_2^2 - x_1^2 - y_1^2 \\ d_1^2 - d_3^2 + x_3^2 + y_3^2 - x_1^2 - y_1^2 \\ d_1^2 - d_4^2 + x_4^2 + y_4^2 - x_1^2 - y_1^2 \end{bmatrix} \quad (8)$$

The coordinates (x_p, y_p) of the node P can be found by solving the system of equations (8).

III. COMBINING THE RSSI/INS DATA

The location of the nodes obtained using RSSI localization is not accurate since the radio signal may be distorted by some factors which may be found in enclosed spaces. To get the required accuracy of location data obtained on the basis of RSSI information is connected to the location data of the INS module. These two sources of information are characterized by complementary properties. INS data providing positioning and orientation is more accurate in the short term but errors, due to deviations affecting the accelerometer and gyroscopic sensors, begin to increase over time. Unlike INS the distance, calculated with the usage of RSSI information, has limited error in the long term but in the short term has low accuracy.

a. Extended Kalman filter

There are plenty of methods, which can be used for RSSI and INS data integration. As a method of data combining the Extended Kalman filter (EKF) is chosen because this method can be easily used for nonlinear system.

The following is a brief description of the EKF used to assess the state of a discrete dynamic system:

$$x_k = f(x_{k-1}, u_k, w_{k-1}) \\ z_k = h(x_k, v_k) \quad (9)$$

x_k - state vector,
u_k - control input,
w_k - process noise,
z_k - system output,
v_k - measuring noise.

The algorithm consists of two repeating phases: forecasting, where the next state of the system is predicted and correction, where, based on the measurements obtained at the output of the system, amendments are made to the state of the predicted system [8].

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In the "prediction" phase $\hat{x}_k^-$ (a priori assessment) and $\hat{P}_k^-$ (error covariance prediction) can be obtained from the following relations:

$$\hat{x}_k^- = f(\hat{x}_{k-1}, u_k, 0)$$
$$P_k^- = A_k P_{k-1} A_k^T + W_k Q_{k-1} W_k^T \quad (10)$$

A_k - The Jacobi matrix f is relatively to x, determined by the following relation:

$$A_k = \frac{\partial f}{\partial x}(\hat{x}_{k-1}, u_k, 0) \quad (11)$$

W_k - the Jacobi matrix f is relatively to w, determined by the following relation:

$$W_k = \frac{\partial f}{\partial w}(\hat{x}_{k-1}, u_k, 0) \quad (12)$$

Q_k - The noise covariance matrix w_k,
P_k - the noise covariance matrix v_k at the system output.

In the "correction" phase the subsequent state of the system can be calculated using the following relationships:

$$\hat{x}_k = f(\hat{x}_k^-) + K_k(z_k - h(\hat{x}_k^-, 0)) \quad (13)$$

K_k - the Kalman strength, which is determined by the following relationship:

$$K_k = P_k^- H_k^T (H_k P_k^- H_k^T + V_k R_k^- V_k^T) \quad (14)$$

The algorithm of Extended Kalman Filter is presented in Figure 2.

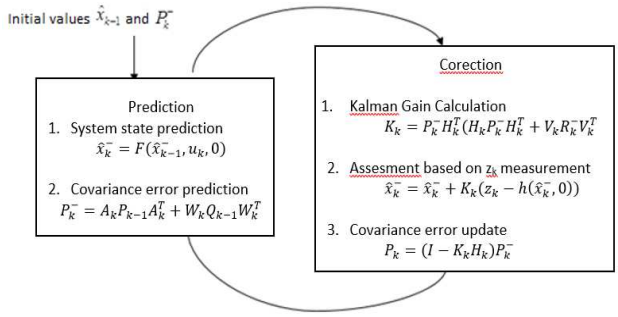


Figure 2. Extended Kalman Filter algorithm

b. Dynamic system model

Consider the state vector of the dynamic model, which is determined by the system coordinates and velocities along the *x, y* axes. The transition of a system from one state to another can be shown as follows:

$$\begin{bmatrix} x \\ y \\ u \\ v \end{bmatrix}_{k+1} = \begin{bmatrix} 1 & 0 & T & 0 \\ 0 & 1 & 0 & T \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ u \\ v \end{bmatrix}_k \quad (15)$$

From the previous section, we know that you can calculate the coordinates (*x, y*) using the RSSI method. For this reason, $(x, y)^T$ is assumed to be the original vector. The initial data of the system will be expressed as the following relationship:

$$z_k = \begin{bmatrix} x \\ u \end{bmatrix}_k = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ u \\ v \end{bmatrix}_k \quad (16)$$

one can easily determine the matrices of passage and measurement of the dynamic system (15) and (16):

$$A_k = \begin{bmatrix} 1 & 0 & T & 0 \\ 0 & 1 & 0 & T \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad H_k = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (17)$$

T- the approximate period.

Suppose the state of an extended Kalman filter means an error in the state vector of a dynamical system. The Kalman filter dynamics is as follows:

$$\begin{bmatrix} \partial x \\ \partial y \\ \partial u \\ \partial v \end{bmatrix}_{k+1} = A_k \begin{bmatrix} \partial x \\ \partial y \\ \partial u \\ \partial v \end{bmatrix}_k \quad (18)$$

Equation (18) indicates that we can assume the error state of the dynamical system a priori.

IV. SOLUTIONS AND RESULTS

The positioning algorithm was carried out using the Arduino Uno mobile node equipped with additional

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INS sensors and the iBeacon stationary node used as a beacon (fixed reference point).

When navigation system implementation four reference points are used - nrf51822 iBeacon (consists of a module and a battery).

The mobile node consists of the following hardware: Arduino Uno microcontroller board, IMU module (inertial measurement unit), and a lithium-polymer 2-cell battery.

During the experiment, the relationship between the RSSI value and the distance of the mobile nodes with fixed reference points was determined. In order to find this relationship, the average RSSI values were measured in several places of the room. The results are shown in table1.

TABLE I. RSSI

Pops(m)	Y_pos(m)	Distance (m)	RSSI
1	1	1.41	-52.05
2	2	2.79	-54.97
3	3	4.24	-57.08
4	4	5.65	-58.81

Using experimental data, the following relationship between RSSI and distance was determined:

$$RSSI(d) = -11.85\log_{10}(d) - 48.82 \quad (19)$$

In addition, the relationship between the RSSI value and the distance can be displayed as a graph in Figure 3.

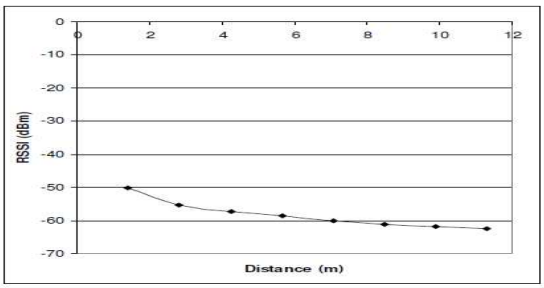


Figure 3. Relationship between distance and RSSI values

The data merging algorithm has been tested in the Matlab Simulink environment. Mobile nodes were considered movable on a plane with constant acceleration. The speed and state of the object's movement are calculated using the following differential equations:

$$\dot{u} = a_x \quad \dot{x} = v$$
$$\dot{v} = a_x \quad \dot{y} = u \quad (20)$$

For accelerometer, used in INS, the following measurement ratios are provided:

$$a_{x_imu} = a_x + w_{acc}$$
$$\dot{u} = a_x \quad (21)$$

w_{acc}- the white noise of a sensor with dispersion $\sigma^2=2mg$

It was taken into account that the location, which was determined by RSSI indicators, as described in equation (22), is affected by the white noise with a dispersion of 2 m and a zero-average value.

$$x_{RSSI} = x + w_{RSSI}$$
$$y = y + w_{RSSI} \quad (22)$$

Figure 4 shows a block diagram of Simulink, a Kalman filter, combining data obtained from RSSI and IMU submodules.

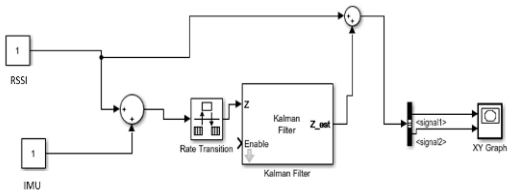


Figure 4. Usage of Kalman filter in the MATLAB SIMULINK environment

Figure 5 shows that algorithm of localization, implemented using only RSSI values, has low accuracy; max distance between calculated area and exact one is 4 meters.

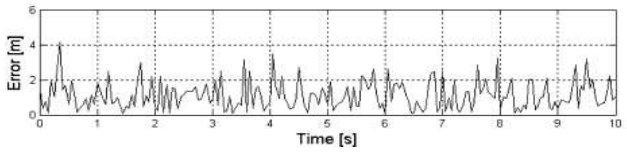


Figure 5. RSSI error assessment for location

The result of proposed location algorithm can be seen in Figure 6. The accuracy of calculated position is increased by using two sources of information (RSSI and INS).

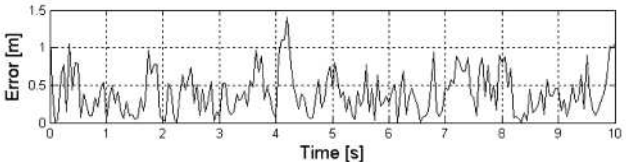


Figure 6. Error assessment of proposed system for location

V. CONCLUSION

Positioning based on received signal strength (RSSI) is the main method for locating objects in wireless sensor networks. However, RSSI-based methods are of low accuracy. This article describes

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navigation systems based on Bluetooth and INS, presents the results of experiments on determining the location of a mobile node using signal strength indicators of a Bluetooth device. The Extended Kalman filter was used to improve positioning accuracy. The accuracy of location determination can be seen from Figure 6, max distance between calculated area and exact one is 1,35 meters.

REFERENCES

[1] Kereyev A.K., Atanov S.K, Aman K.P., Kulmagambetova Zh.K., Kulzhagarova B.T. (2020). Navigation system based on Bluetooth beacons: implementation and experimental estimation, *Journal of Theoretical and Applied Information Technology*, 98(8),1187-1200

[2] Faragher R., Robert H. (2015) Location fingerprinting with bluetooth low energy beacons, *IEEE Journal on Selected Areas in Communications,* 33, 2418–2428.

[3] Rizos, Chris, *Trends in GPS Technology & Applications*, 2021

[4] Zhao X., Xiao Z., Markham A., Trigoni N., Ren Y.(2014). Does BTLE measure up against wifi? A comparison of indoor location performance, *In Proceedings of the European Wireless 2014 20th European Wireless Conference* (pp. 263–268). Barcelona

[5] Danis, F.S. Cemgil, A.T. (2017). Based Localization and Tracking Using Bluetooth Low–Energy beacons, *Sensors,* 17, 2484.

[6] Cabarkapa D., Grujic I., Pavlovic P. (2015) Comparative Analysis of the Bluetooth Low–Energy Indoor Positioning systems, *In Proceedings of the 12th International Conference on Telecommunication in Modern Satellite, Cable and Broadcasting Services (TELSIKS)* (pp. 76–79), Nis, Serbia.

[7] Neburka J., Tlamsa Z., Benes V., Polak L., Kaller O., Bolecek L., Sebesta J., Kratochvil T. (2016). Study of the Performance of RSSI Based Bluetooth Smart Indoor Positioning, *In Proceedings of the 26th International Conference Radioelektronika (RADIOELEKTRONIKA)* (pp. 121–125), Kosice, Slovakia.

[8] Bu Y., Seo K., Huh J. (2017). A study of enhancement of ranging performance of beacons through improvement of the smart phone's gyroscope: Focusing on the Bluetooth low energy

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